

# Zhishan Yuan

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## Education

### Stevens Institute of Technology, Hoboken, NJ

*Ph.D. in Computer Science · Advisor: Prof. Yue Ning*

Sep 2025 – Present

GPA: 3.62/4.0

Coursework: Machine Learning (A), Natural Language Processing (A), Knowledge Discovery & Data Mining (A).

### Huazhong University of Science and Technology, Wuhan, China

*B.Eng. in Computer Science*

Sep 2021 – Jun 2025

GPA: 88.4/100

Coursework: Calculus (98), Data Structures & Algorithms (91), Game Theory (90), Linear Algebra (87).

## Publications

### Diagnosis Before Treatment? Investigating Intermediate-Variable Reasoning in LLMs Findings of EMNLP 2026

Zhishan Yuan, Tiago Almeida, Shudong Hao, Zining Zhu, Nickolas Elias Dawlabani, Ghazia Pervaiz, Yue Ning.

- Introduced a controlled phenotype–disease–drug benchmark (719 KG-derived chains from PrimeKG) that hides the intermediate diagnosis, enabling causal interventions on a domain-aligned reasoning variable across **6 open-weight 7B–8B LLMs**.
- Showed diagnosis is behaviorally consequential: supplying the gold diagnosis raised treatment accuracy from **49.9% to 84.9%**, while incorrect diagnoses dropped it to 23.1%; a shuffled-diagnosis control helped separate diagnosis correctness from the presence of extra disease-like text.
- Demonstrated held-out QLoRA path supervision improves treatment selection by **+19.4 pts**, with external validation on HPO–CTD and OptimusKG and layer-wise activation-patching / Center-of-Gravity localization.

## Research Experience

### Graduate Research Assistant — Stevens Institute of Technology

*Advisor: Prof. Yue Ning · LLM interpretability & evaluation*

Sep 2025 – Present

- Build reproducible LLM-evaluation pipelines (Python, PyTorch, Hugging Face, JSONL, Slurm/HPC) with teacher-forced / continuation-likelihood scoring and bootstrap confidence intervals.
- Run QLoRA/LoRA supervision experiments across general and biomedical LLMs, comparing diagnosis-only, treatment-only, full-chain, and multitask objectives under held-out splits.
- Probe transformer internals via activation patching, SVD subspace analysis, and layer-wise localization.

### Graduate Teaching Assistant — Stevens Institute of Technology

*Machine learning / programming / data analysis*

Fall 2026

## Selected Projects

### Diagnosis-Related Causal Subspace Analysis

*Ongoing*

- Developing hidden-state patching and subspace-isolation methods to localize diagnosis-aligned information in LLM representations and quantify its causal effect on downstream treatment selection.

### Open-Source Biomedical Evaluation Toolkit

*2026*

- Developed benchmark-construction and evaluation code: distractor filtering, split-overlap auditing, option-only baselines, prompt templates, and result aggregation for reproducible LLM analysis.

## Technical Skills

**Programming:** Python, PyTorch, Hugging Face Transformers, NumPy, pandas, scikit-learn, C/C++, MATLAB

**ML / NLP:** LLM evaluation, prompt design, teacher-forced scoring, QLoRA/LoRA, KG benchmarks, medical NLP

**Interpretability:** activation patching, hidden-state & SVD subspace analysis, CoG localization, bootstrap significance

**Systems / Tools:** Linux, Git, Slurm/HPC, CUDA, LaTeX/Overleaf, JSON/JSONL pipelines

## Awards & Honors

- **Outstanding Undergraduate for Academic Performance**, HUST — top 1% by academic performance (Dec 2022).
- **Imperial College London Summer School** — Distinction (Grade A).